Inverse Problems in Geophysics
Part 7: Regularization

2. MGPY+MGIN

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Recap Regularization

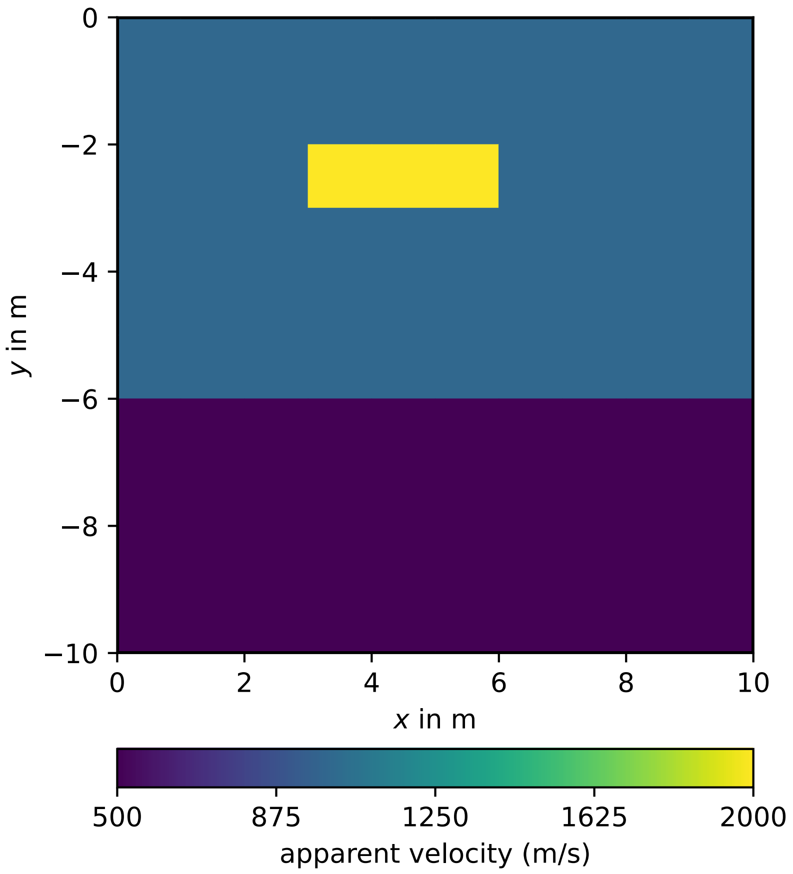
Ill-condition: large vs. small low singular values (amplifying noise)

1. TSVD: Truncating matrix approximation at pseudo-rank
2. Damped normal equations (minimum norm) through filter factors
3. Smoothness constraints explicitly minimize smooth models

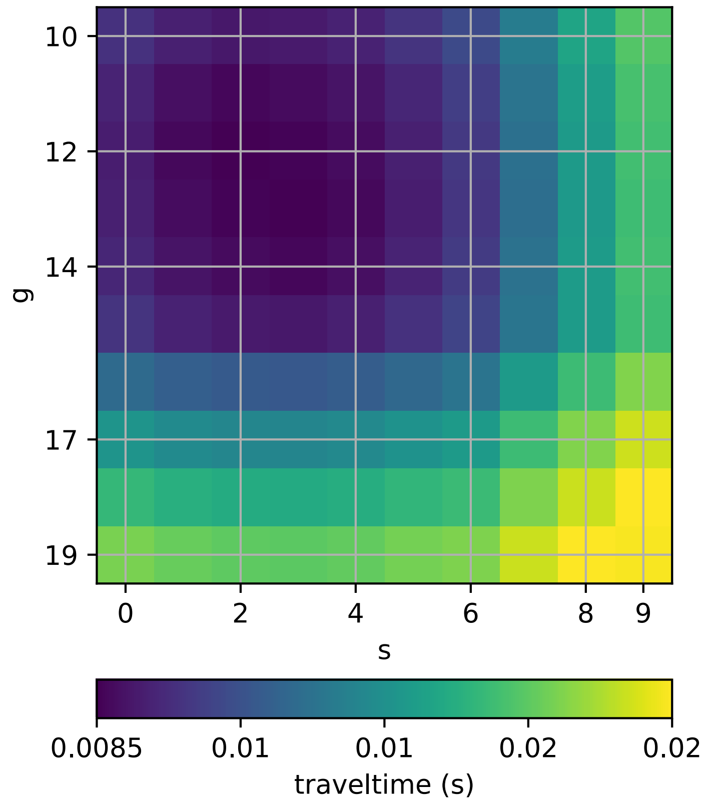
- choice of regularization strength (λ, p) is vital
  - watch model (roughness) & data misfit (plot), discrepancy ($\chi^2 = 1$)
- TT resolution: best in the center, non-horizontal rays important

A synthetic model

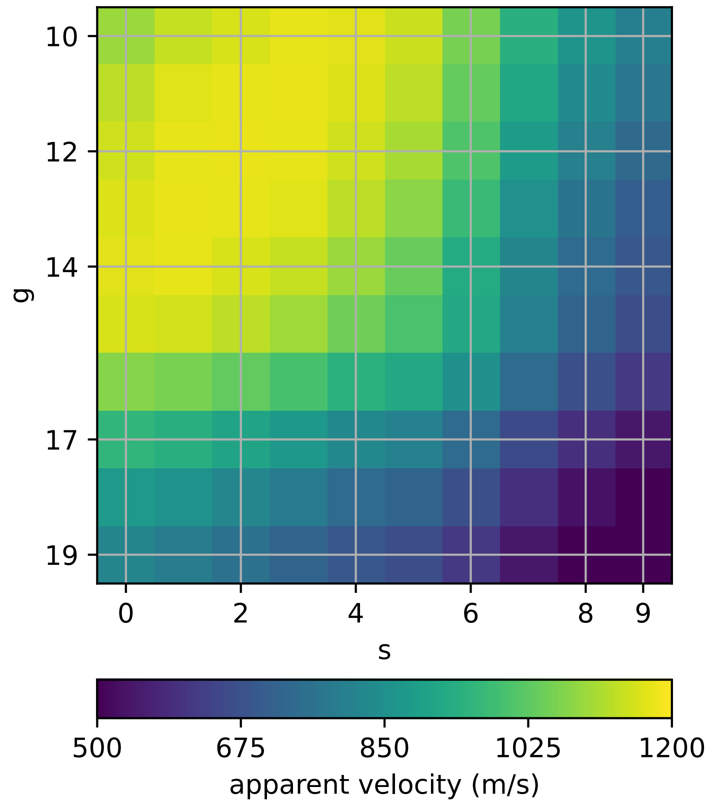
Synthetic model



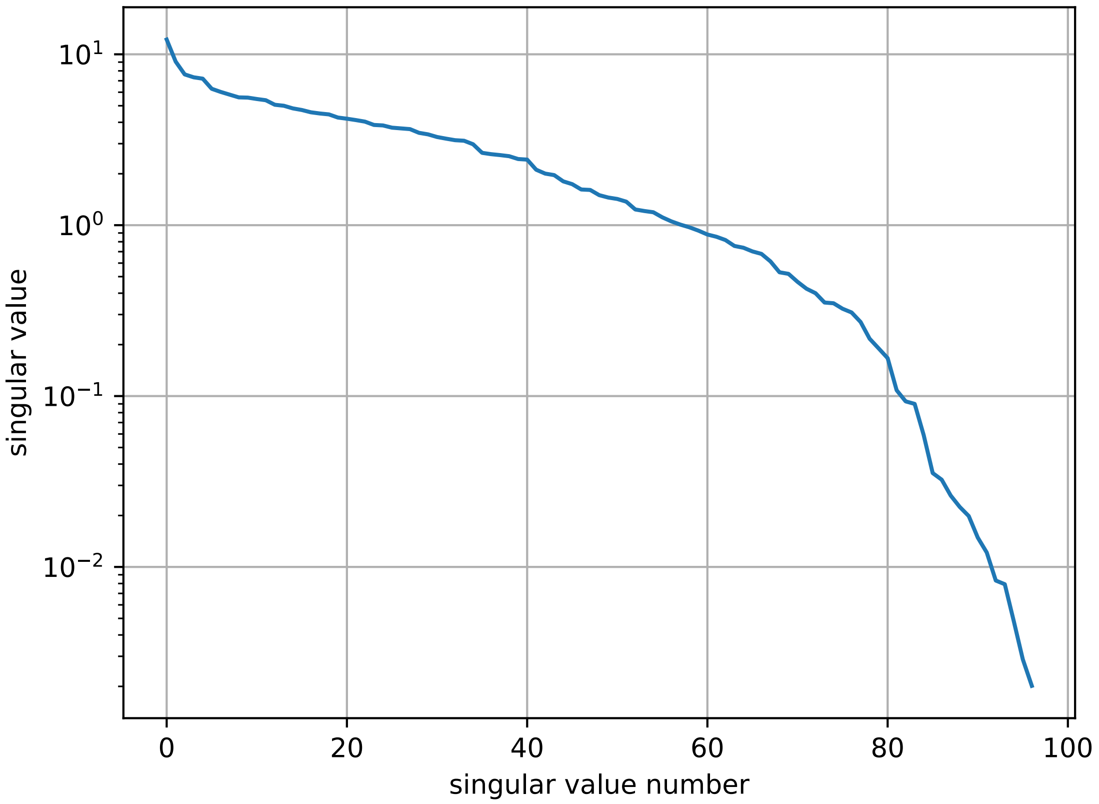
Traveltimes



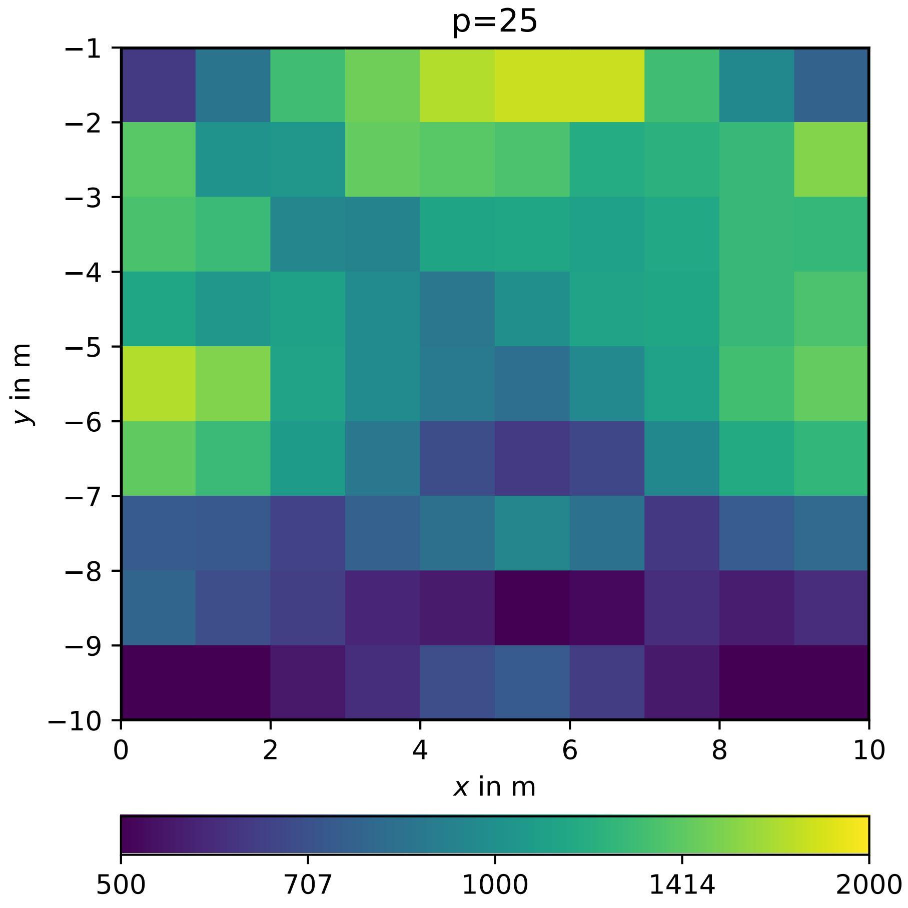
Apparent velocity



TSVD result

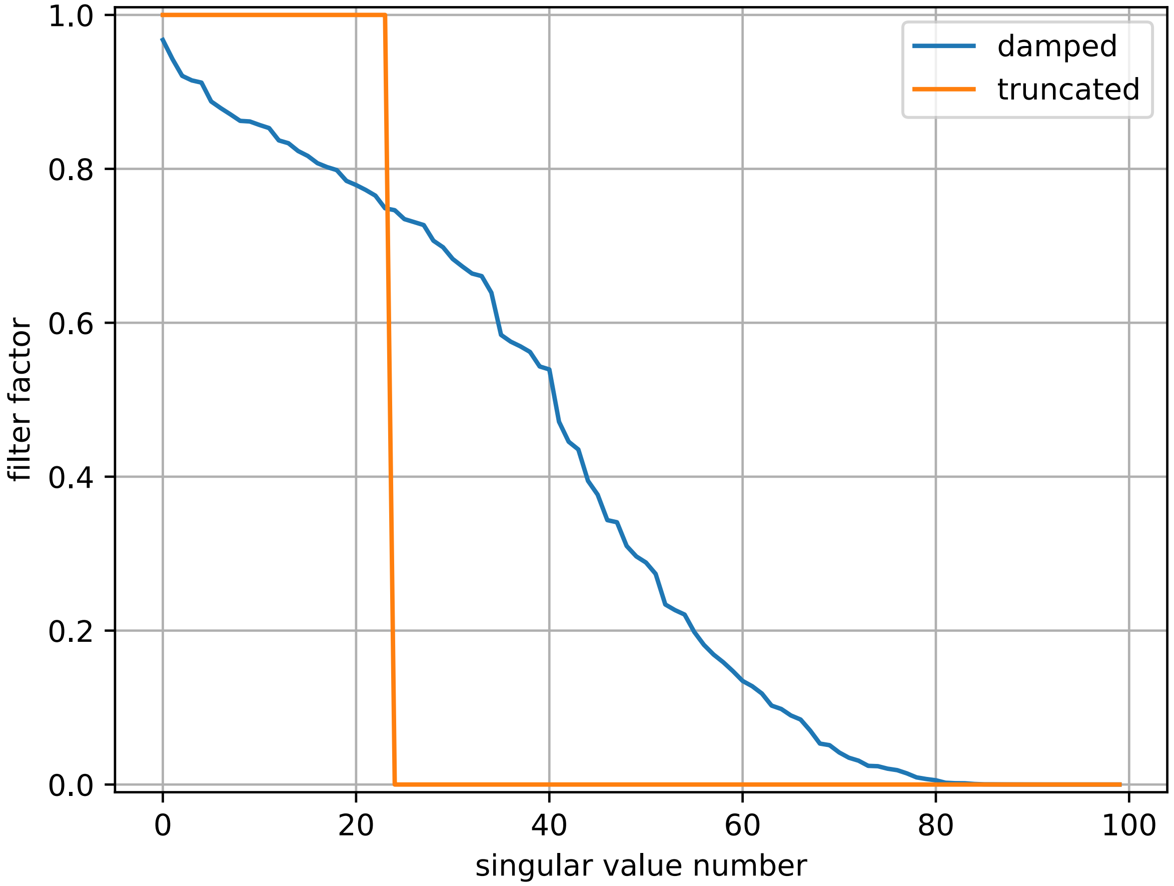


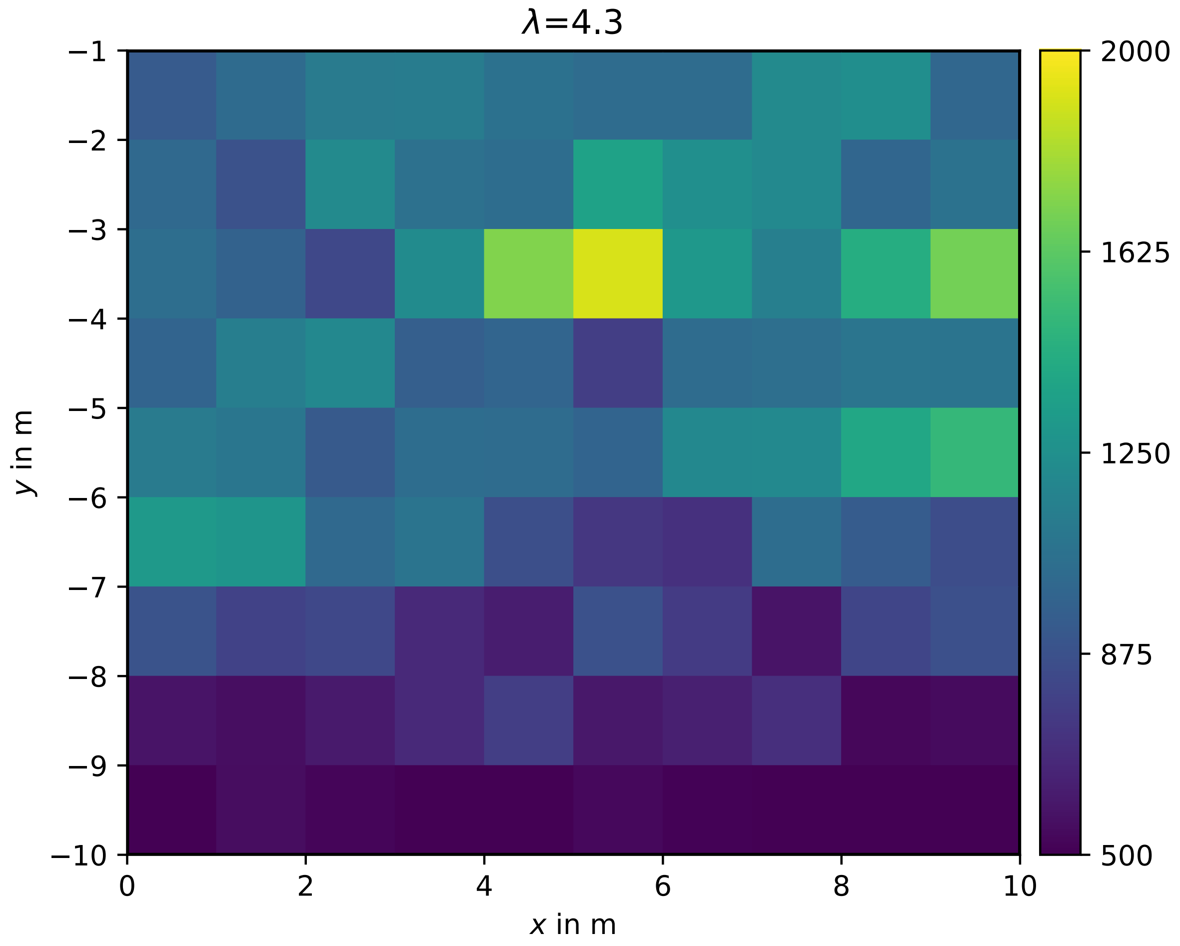
10^1
10^0
10^{-1}
10^{-2}
singular value
0
20
40
60
80
100
singular value number



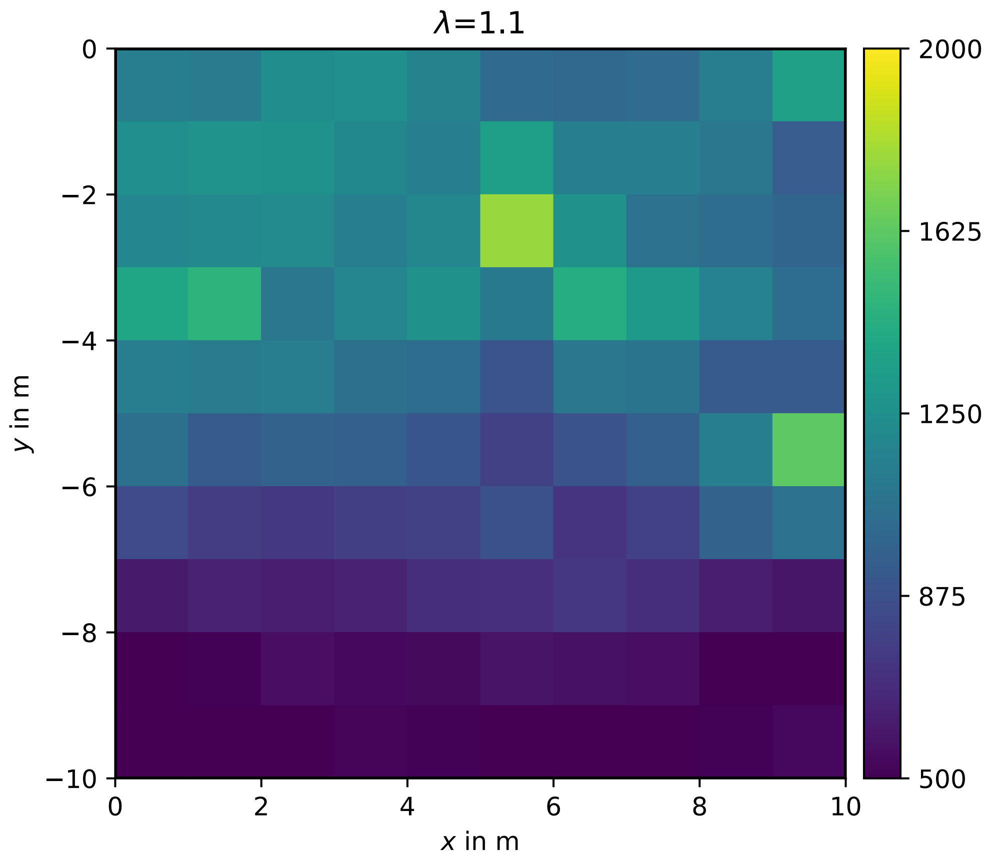
p=25
−1
−2
−3
−4
−5
−6
−7
−8
−9
−10
y in m
0
2
4
6
8
10
x in m
500
707
1000
1414
2000

Damped normal equation result

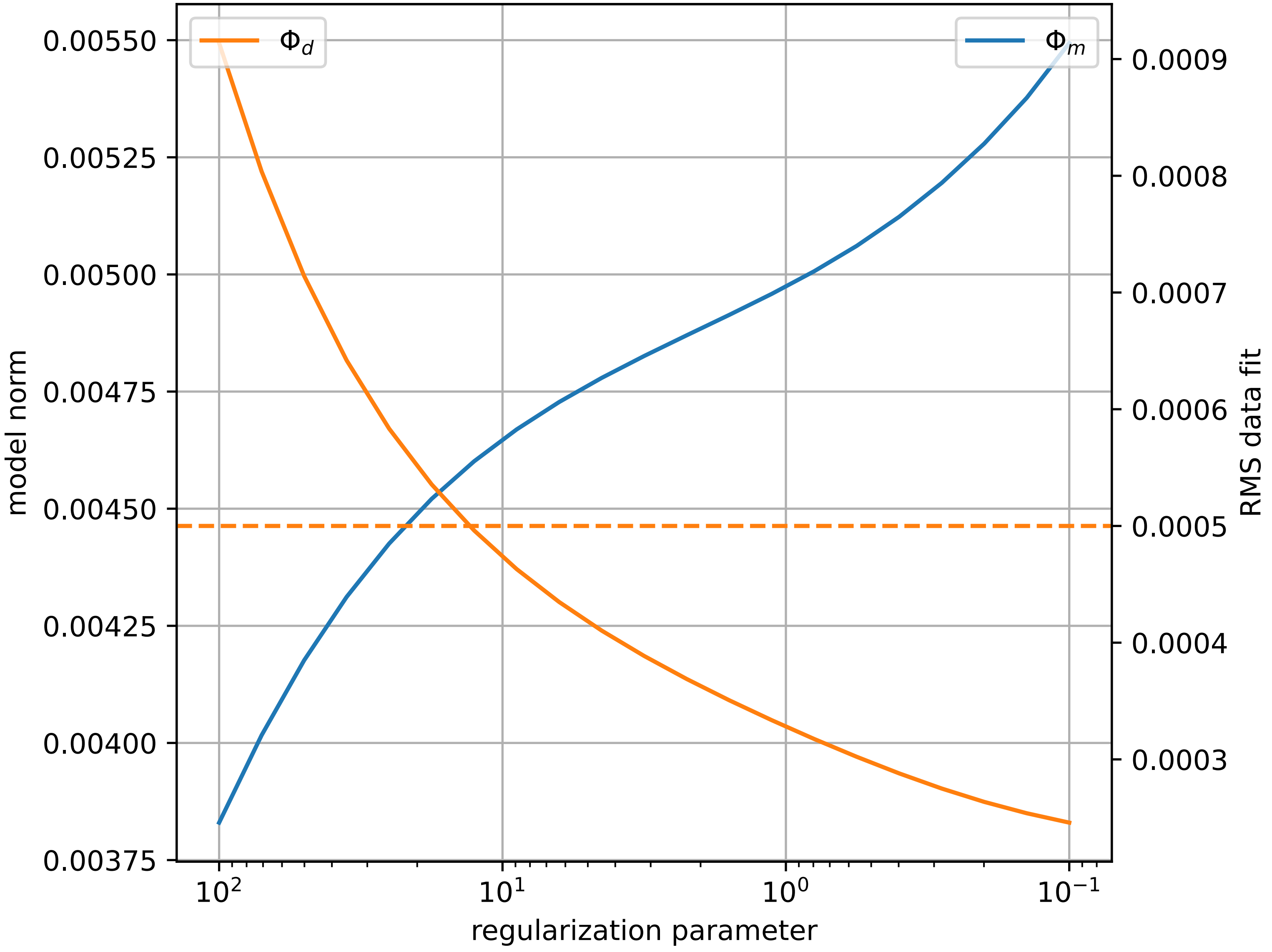




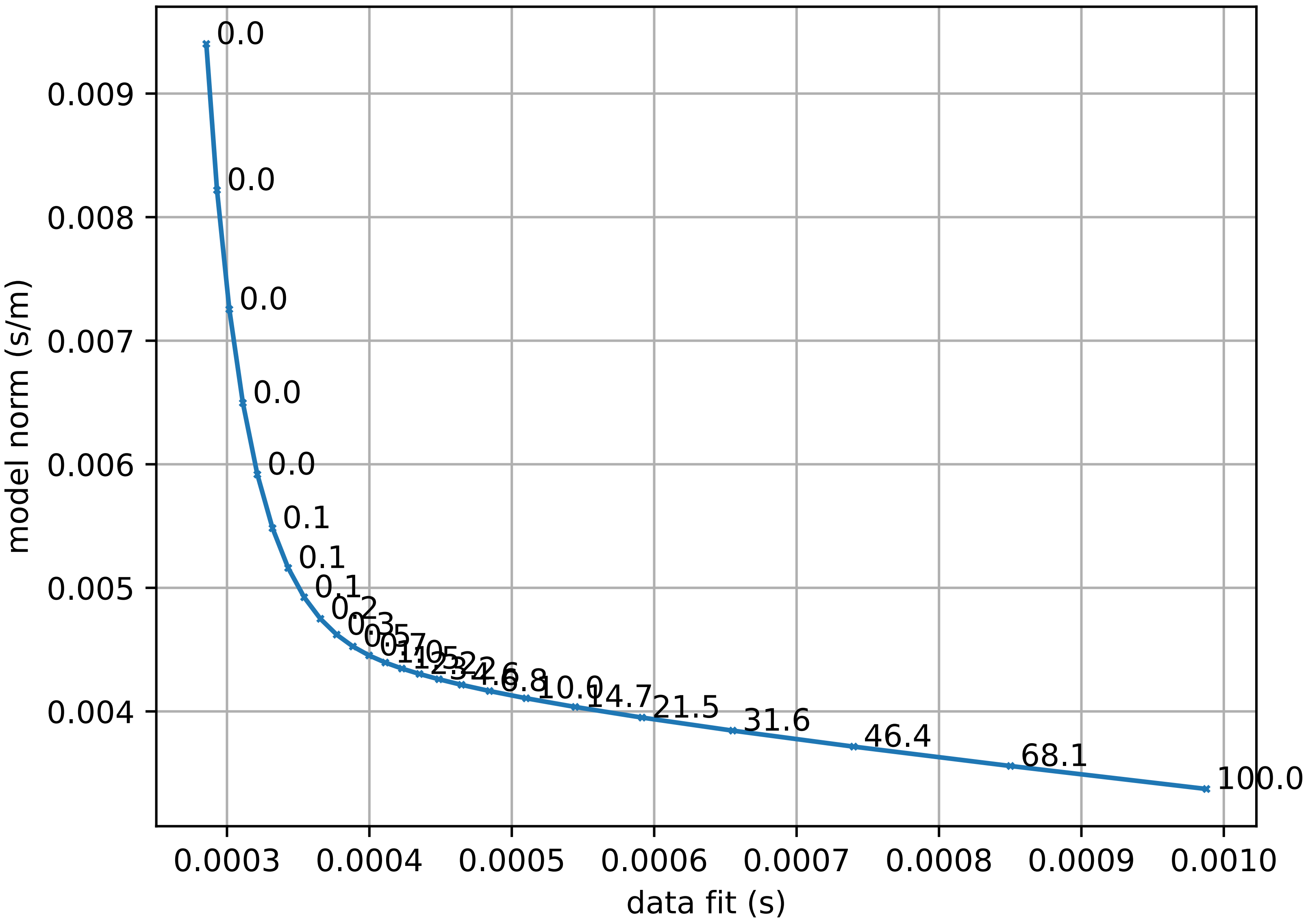
Inversion with smoothness constraints



Smoothness constraints: Data and model norm



Smoothness constraints: The L-curve



Resolution for smoothness constraints

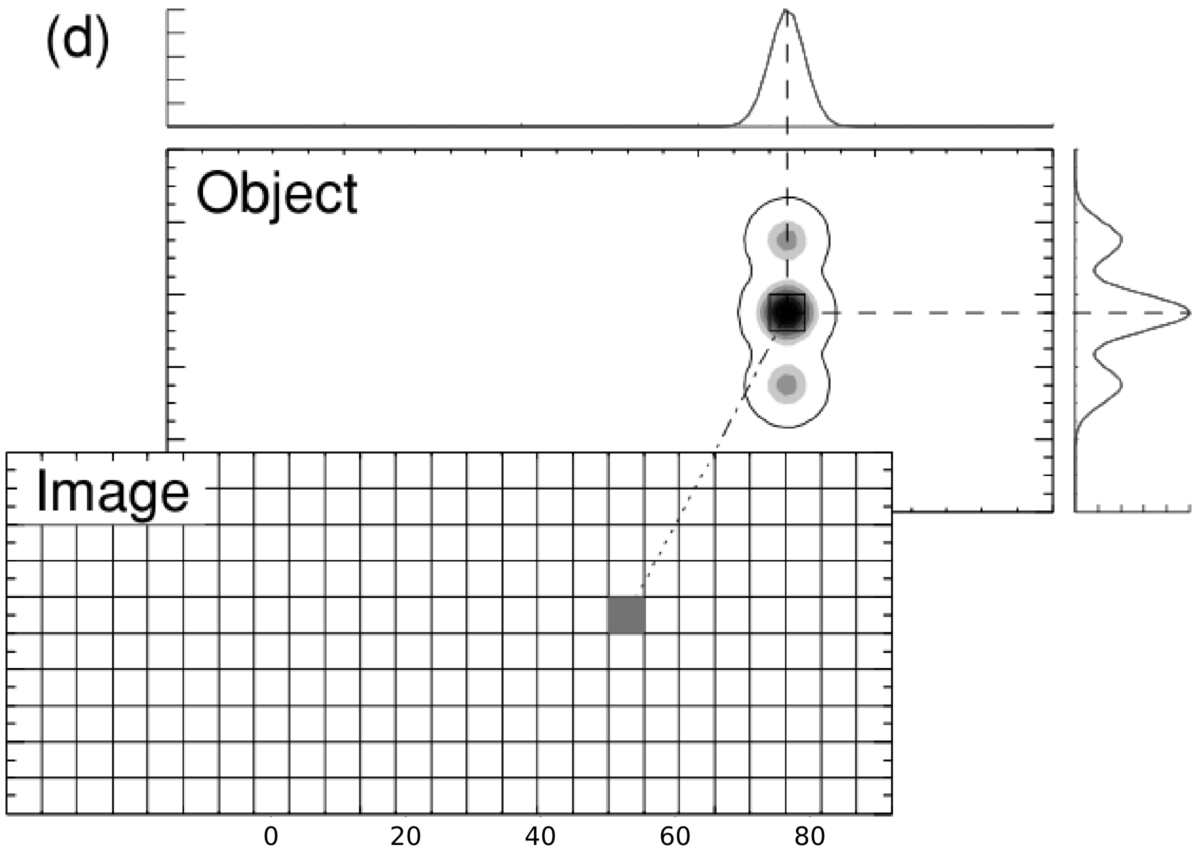
$$\Rightarrow \mathbf{R}^M = \mathbf{G}^\dagger \mathbf{G} =$$

$$(\mathbf{G}^T\mathbf{G} + \lambda \mathbf{C}^T\mathbf{C})^{-1}\mathbf{G}^T\mathbf{G}$$

smoothed across neighboring cells

A column represents how a cell anomaly is "spread" (Friedel 2003)

point spread function (PSF)

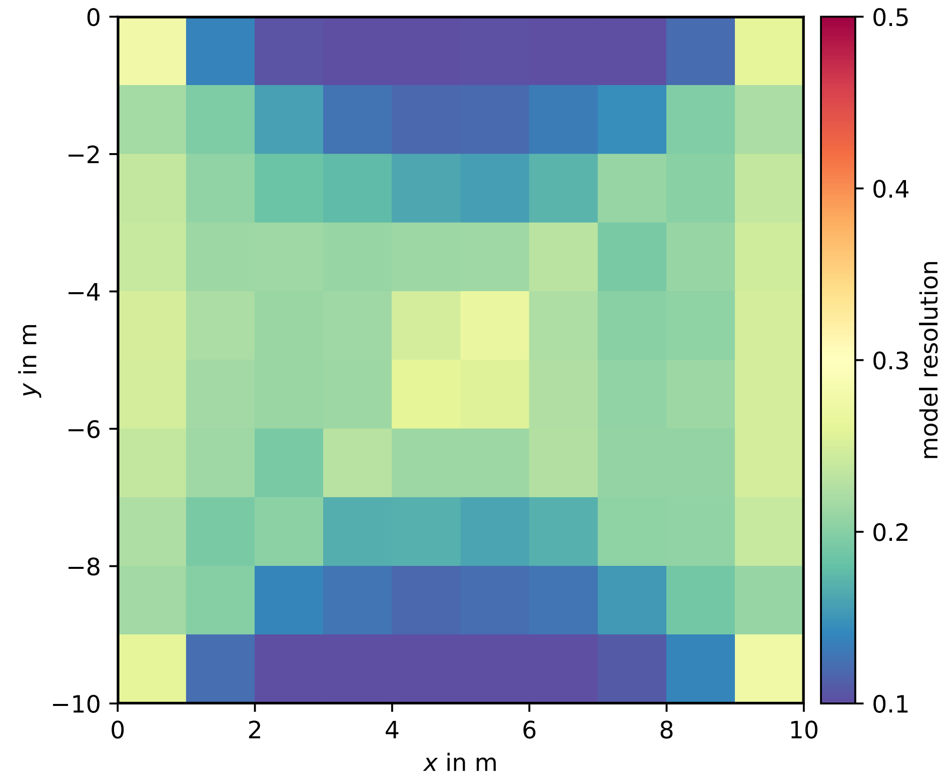


ideal – deficiency – distortion – side lobes

Model resolution

The diagonal of the resolution matrix

states how a single model cell can be retrieved



- far from ideal value (1)
- low at top and bottom center (only horizontal rays)
- highest in the middle and close to sensors

model cell resolution

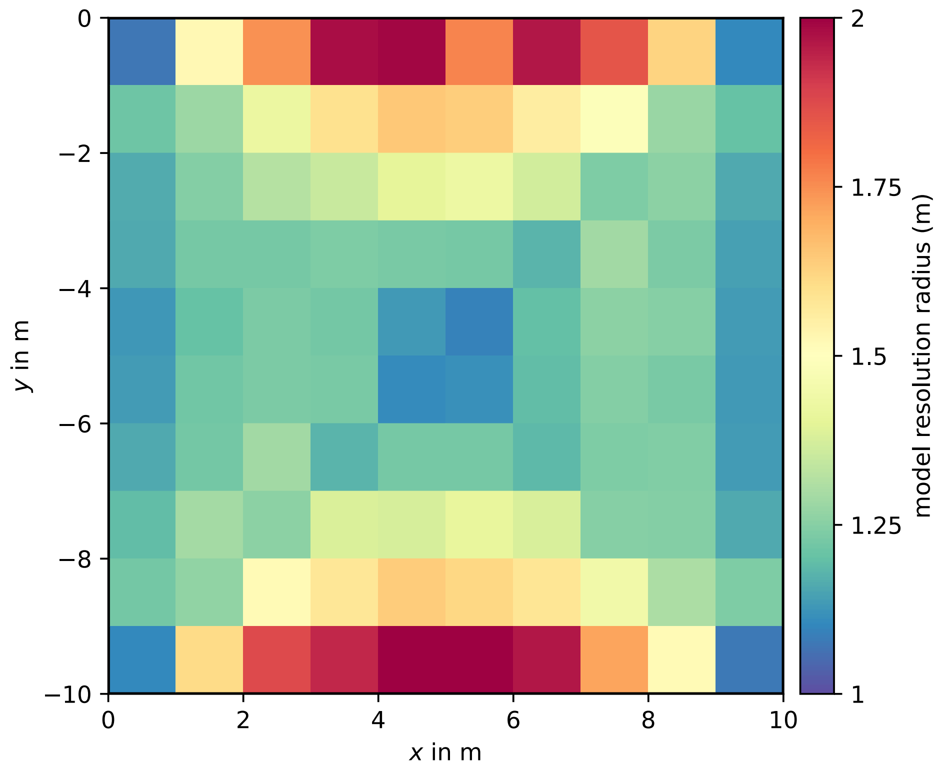
Resolution radius

How big needs a sphere to be to integrate a value of 1 (perfect resolution)?

Resolution radius (Friedel 2003)

$$r = \frac{r_i}{\sqrt{\mathbf{R}_{ii}^M}} = \frac{\sqrt{A_i/\pi}}{\sqrt{\mathbf{R}_{ii}^M}}$$

mesh-independent geometric uncertainty



model cell resolution

Summary

- TSVD & damped normal equations help, but struggle image small bodies
- smoothness constraints provide blurred image, but is robust
- understand smoothing through model resolution matrix (PSFs)
- resolution radius provides geometric uncertainty
- larger regularization (due to larger errors) in worse resolution
- parameter uncertainty not well described by $\mathbf{R}^M$

Model covariance matrix

Let $\mathbf{x}$ be a multivariate, normal distributed random vector with mean μ and covariance $\mathbf{C}$. $\mathbf{y} = \mathbf{A}\mathbf{x}$ is also a multivariate random vector with the expectancy $E(\mathbf{y}) = \mathbf{A}\mu$ and the covariance $\mathrm{cov}(\mathbf{y}) = \mathbf{A}\mathbf{C}\mathbf{A}^T$.

Inverse problem:

$$E(\mathbf{m}) = E(\mathbf{G}^\dagger \mathbf{d}) = \mathbf{G}^\dagger E(\mathbf{d}) = \mathbf{R}^M \mathbf{m}^{true}$$

$$\mathrm{cov}(\mathbf{m}) = \mathbf{G}^\dagger \mathrm{cov}(\mathbf{d}) (\mathbf{G}^\dagger)^T \quad \mathrm{cov}(\mathbf{d}) = \sigma^2$$

$$\Rightarrow \mathrm{cov}(\mathbf{m}) = \sigma^2 (\mathbf{G}^T\mathbf{G})^{-1} \mathbf{G}^T \mathbf{G} (\mathbf{G}^T\mathbf{G})^{-1} = \sigma^2 (\mathbf{G}^T\mathbf{G})^{-1}$$

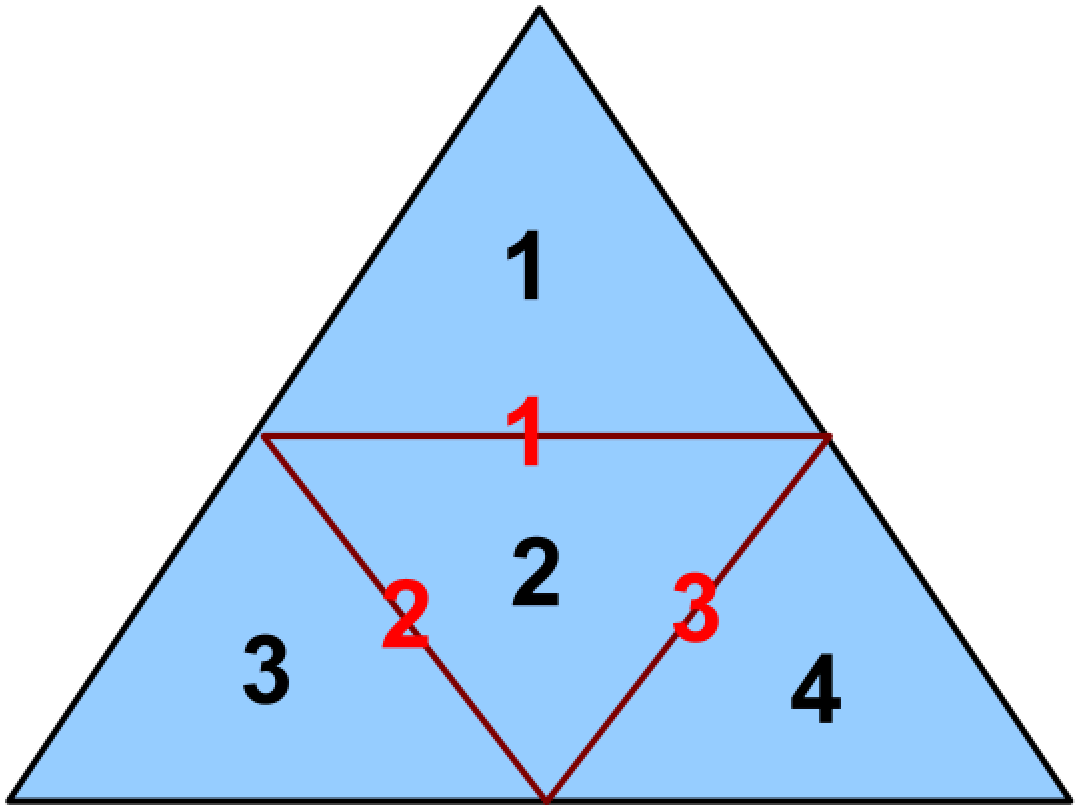
Modern regularization

> ⓘ **Regularization is key to subsurface imaging**

$\Rightarrow$ Add any information about the subsurface:

- correlation lengths and angles
- structural boundaries (from boreholes, seismics, GPR)
- point information (samples, borehole logs,)
- limits of the possible parameters (e.g. positivity, natural bounds)

Smoothness constraints



4 model cells, 3 boundaries

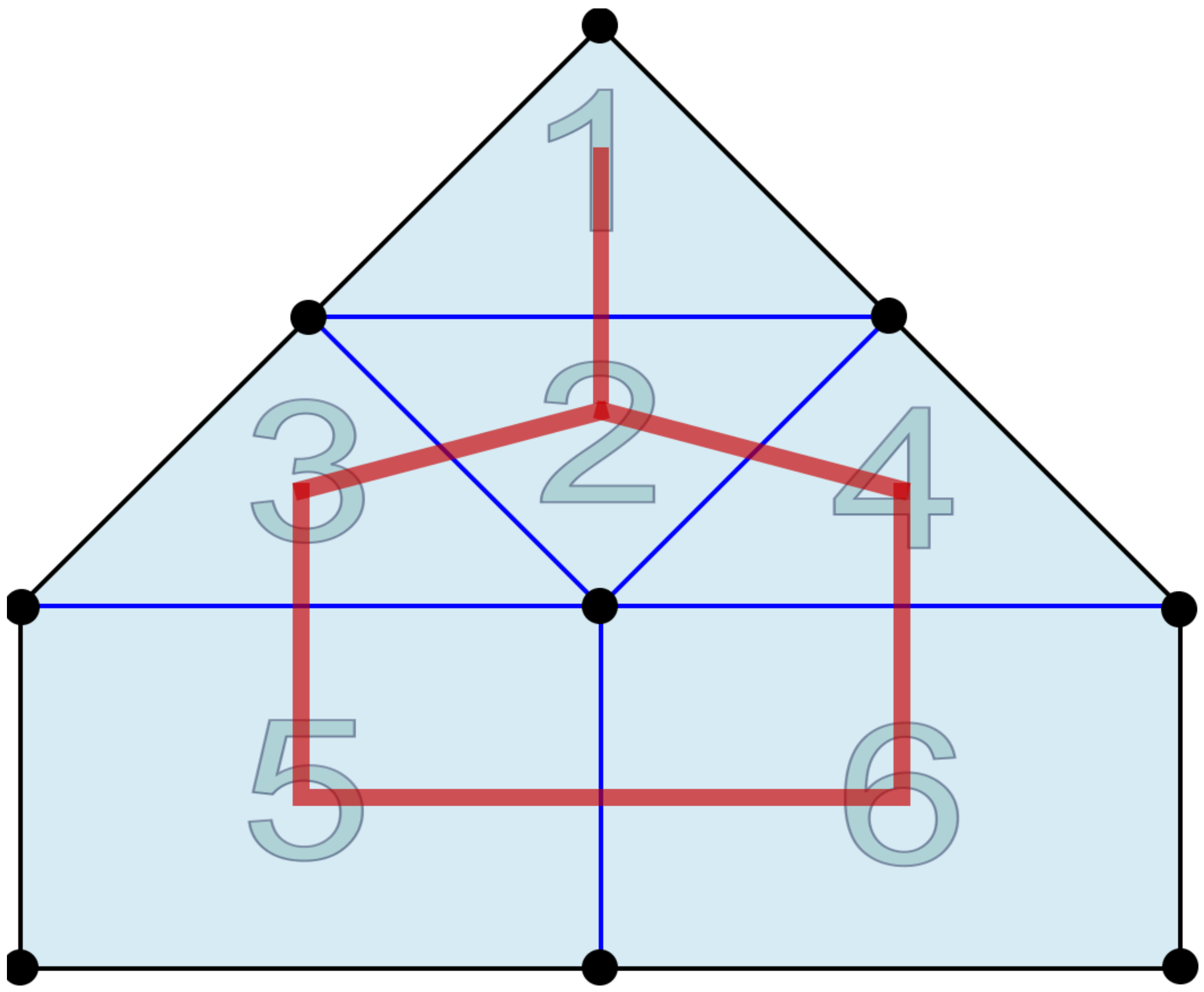
1st order derivative

$$\mathbf{C} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}$$

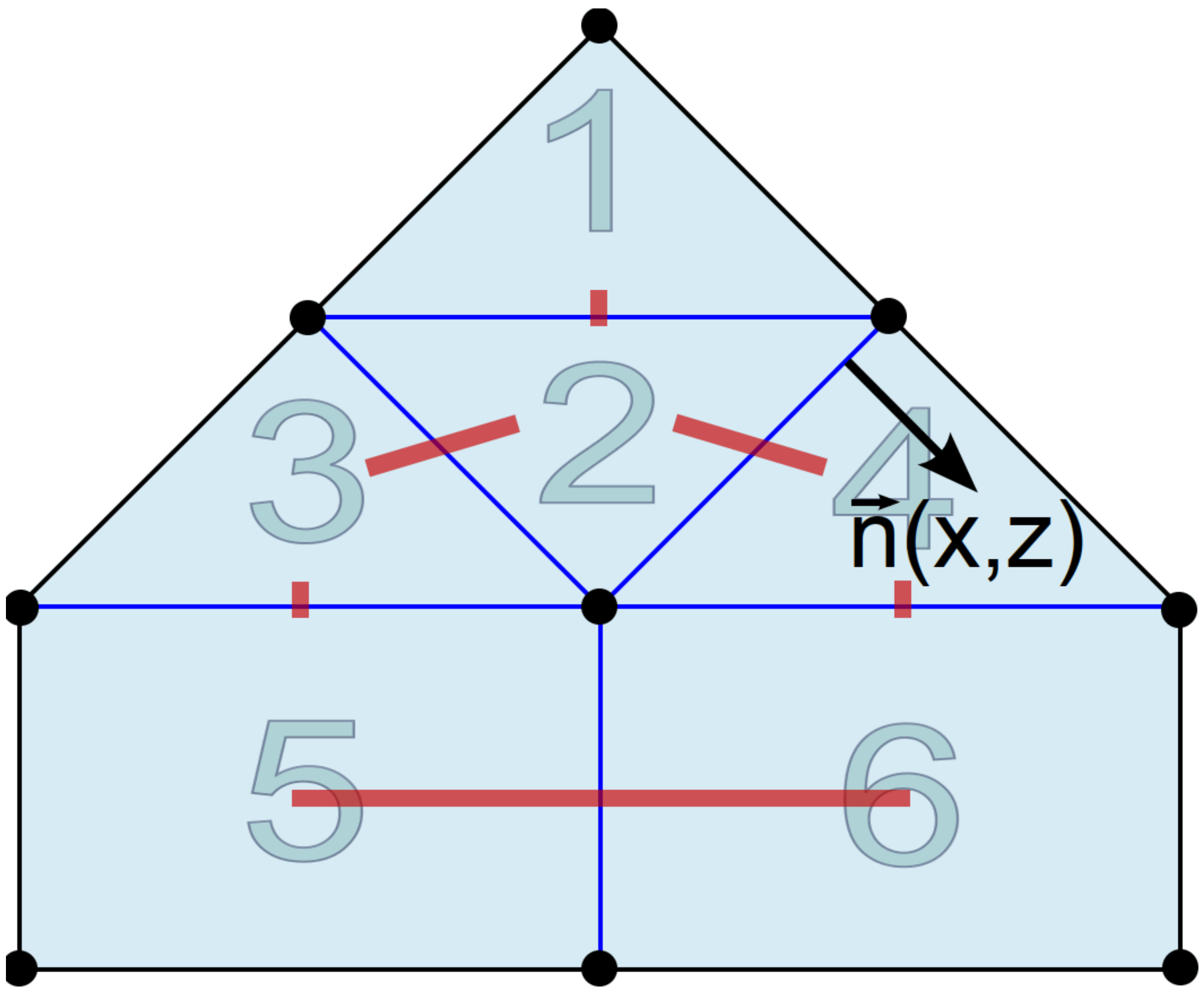
can be weighted by

- inverse midpoint distance
- boundary size (length, area)
- information about boundaries

Anisotropic smoothness

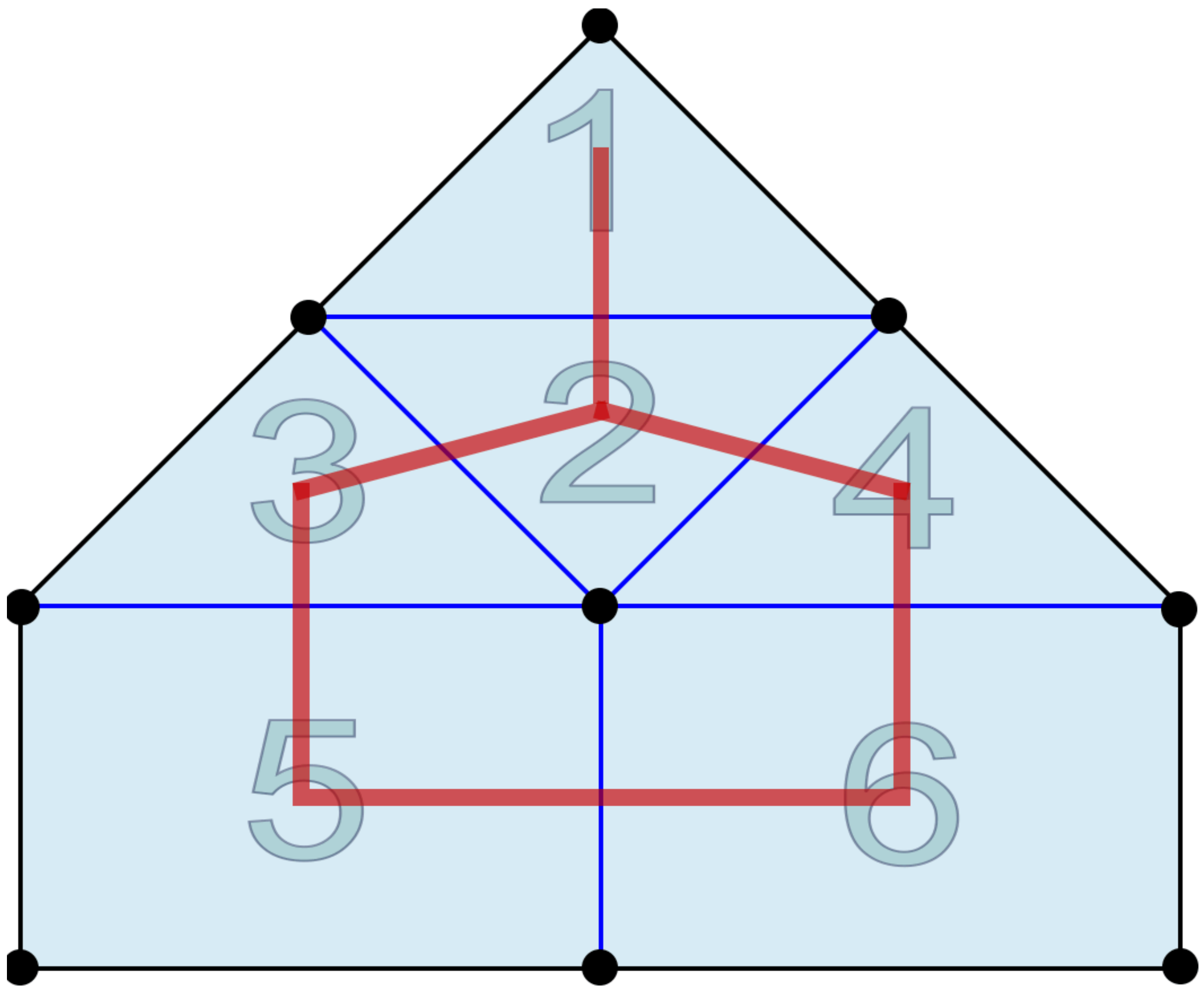


isotropic smoothness

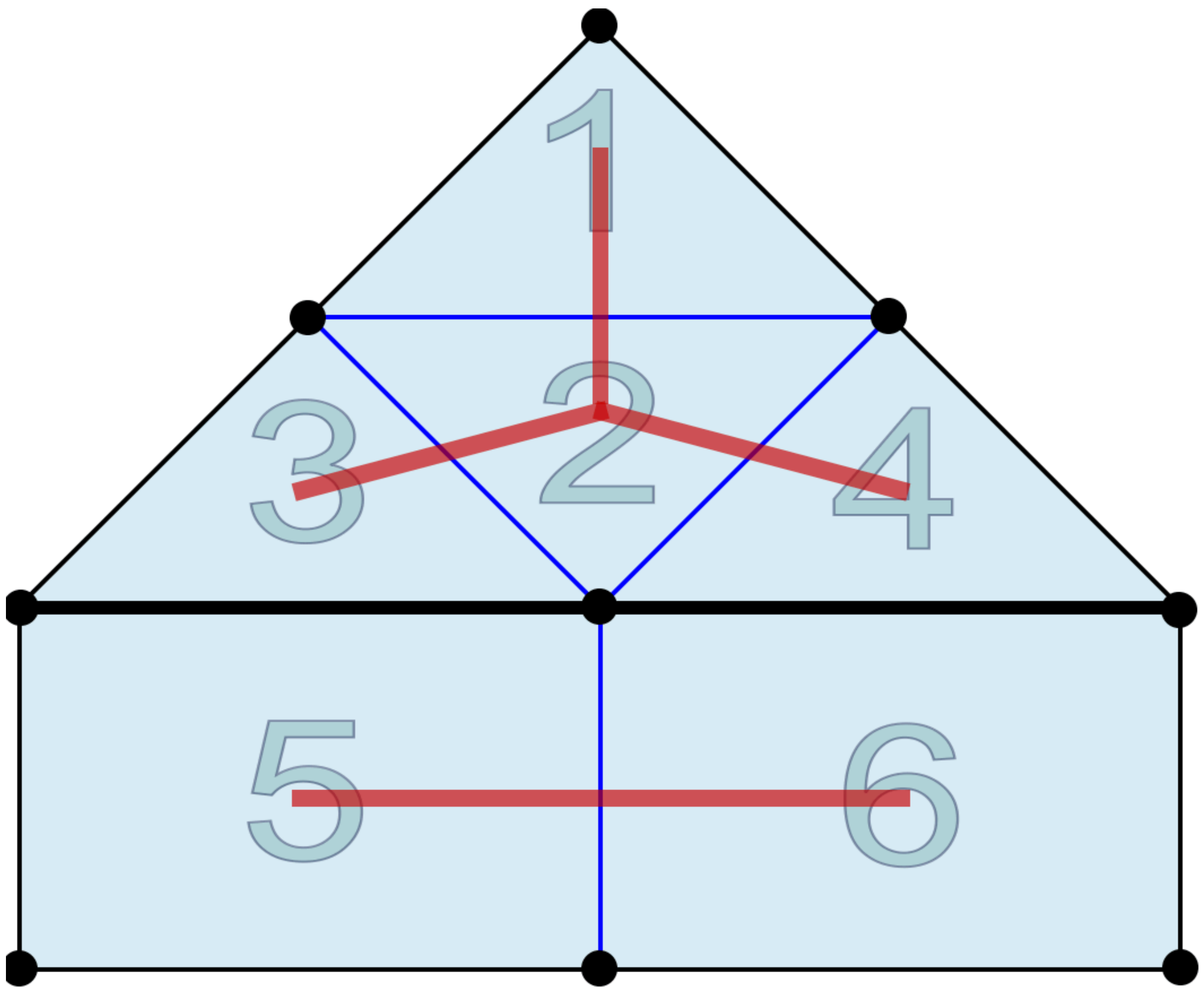


anisotropic smoothness

Structural decoupling

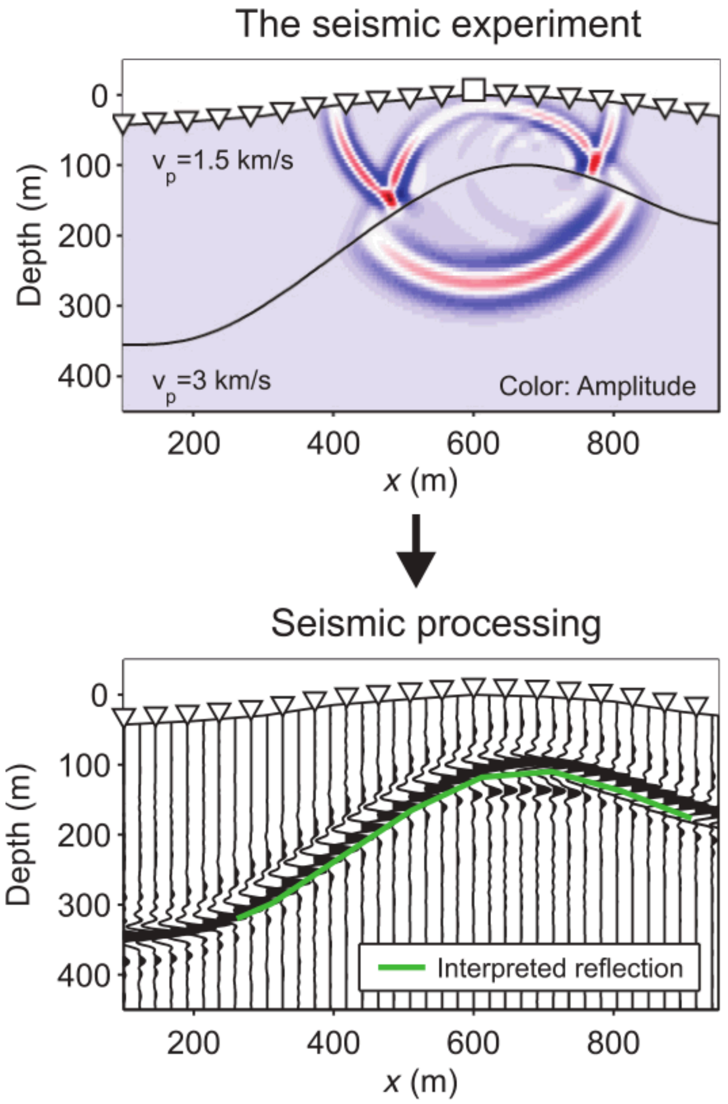


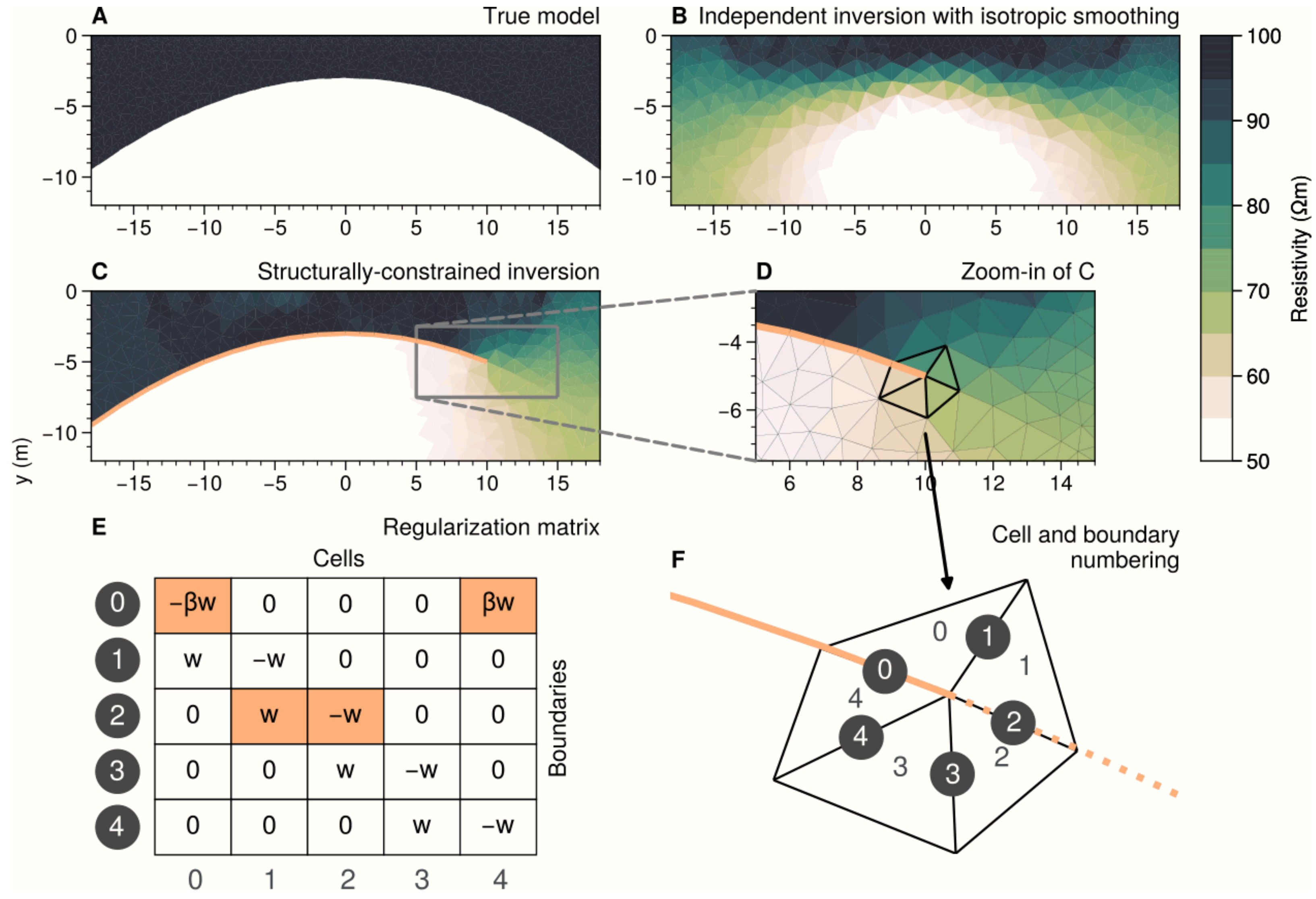
isotropic smoothness



decoupling

Structural constraints



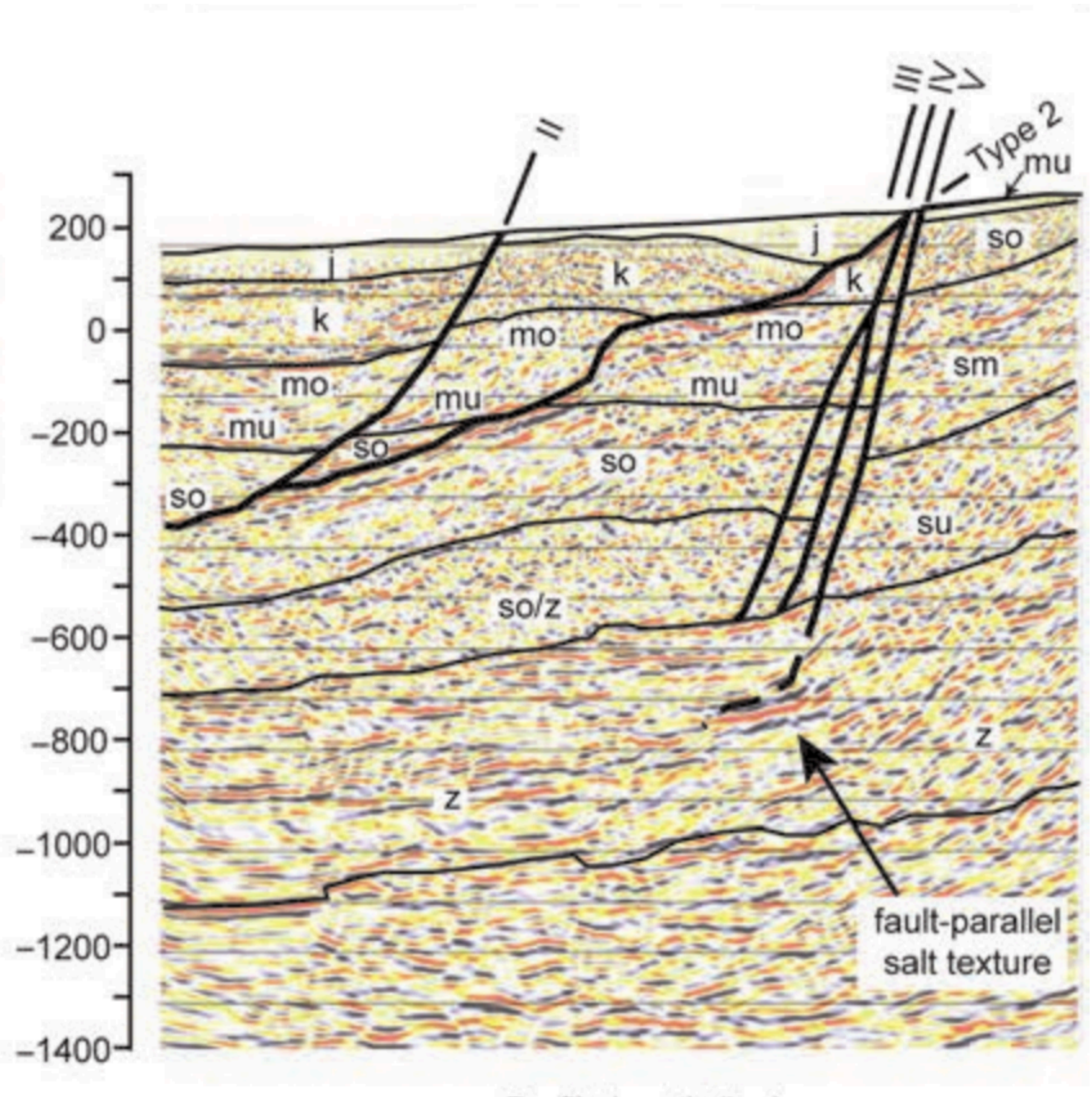


Boundaries \ Cells	0	1	2	3	4
0	−βw	0	0	0	βw
1	w	−w	0	0	0
2	0	w	−w	0	0
3	0	0	w	−w	0
4	0	0	0	w	−w

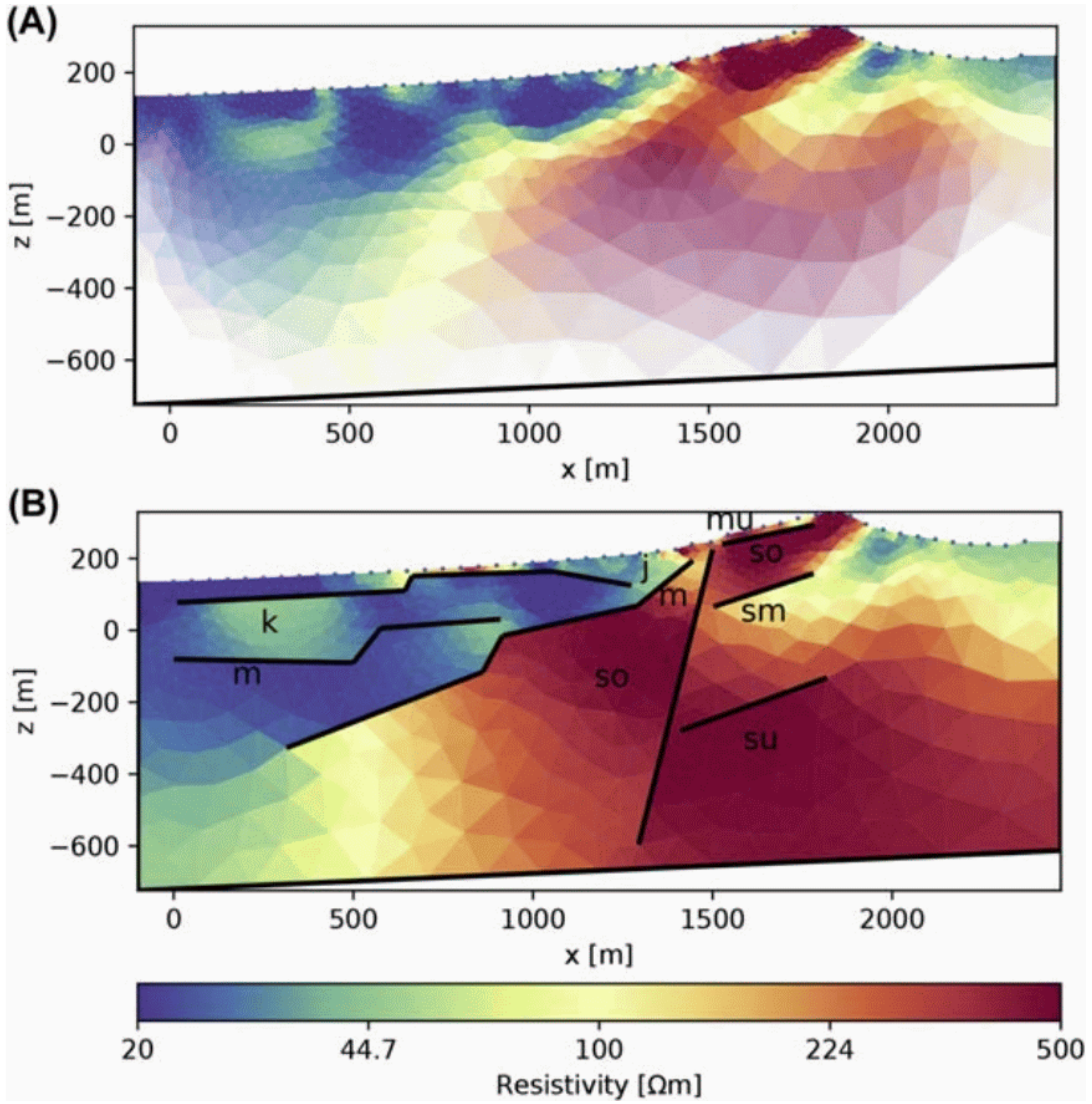
Bergmann et al. 2014

Wagner & Uhlemann 2021

Structural example (Tanner et al. 2020)

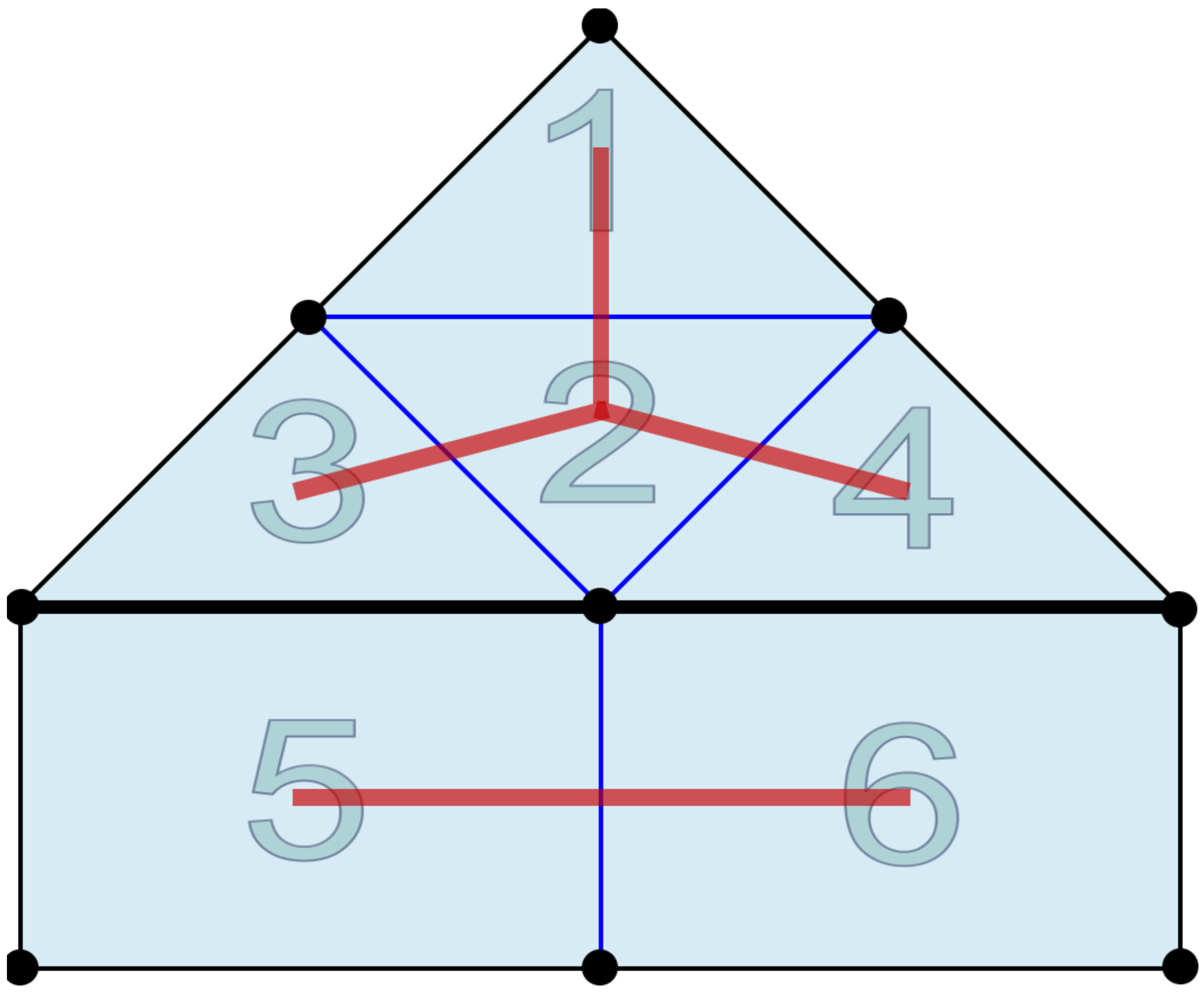


II
III IV V
Type 2
mu
so
j
k
mo
mu
so
sm
su
so/z
z
fault-parallel salt texture
200
0
−200
−400
−600
−800
−1000
−1200
−1400

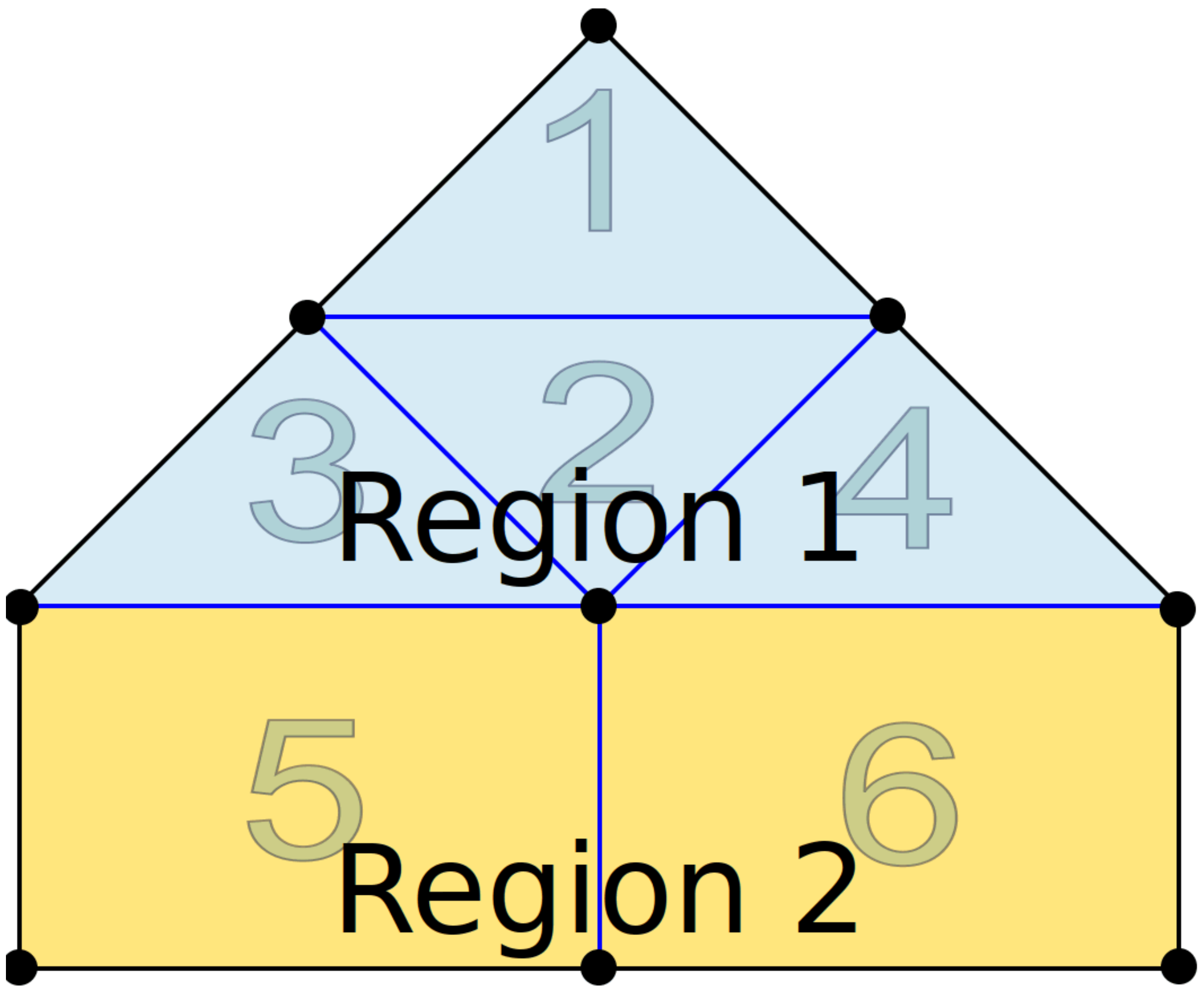


(A)
z [m]
x [m]
(B)
mu
so
j
m
sm
k
m
so
su
Resistivity [Ωm]
20
44.7
100
224
500

Selected smoothness

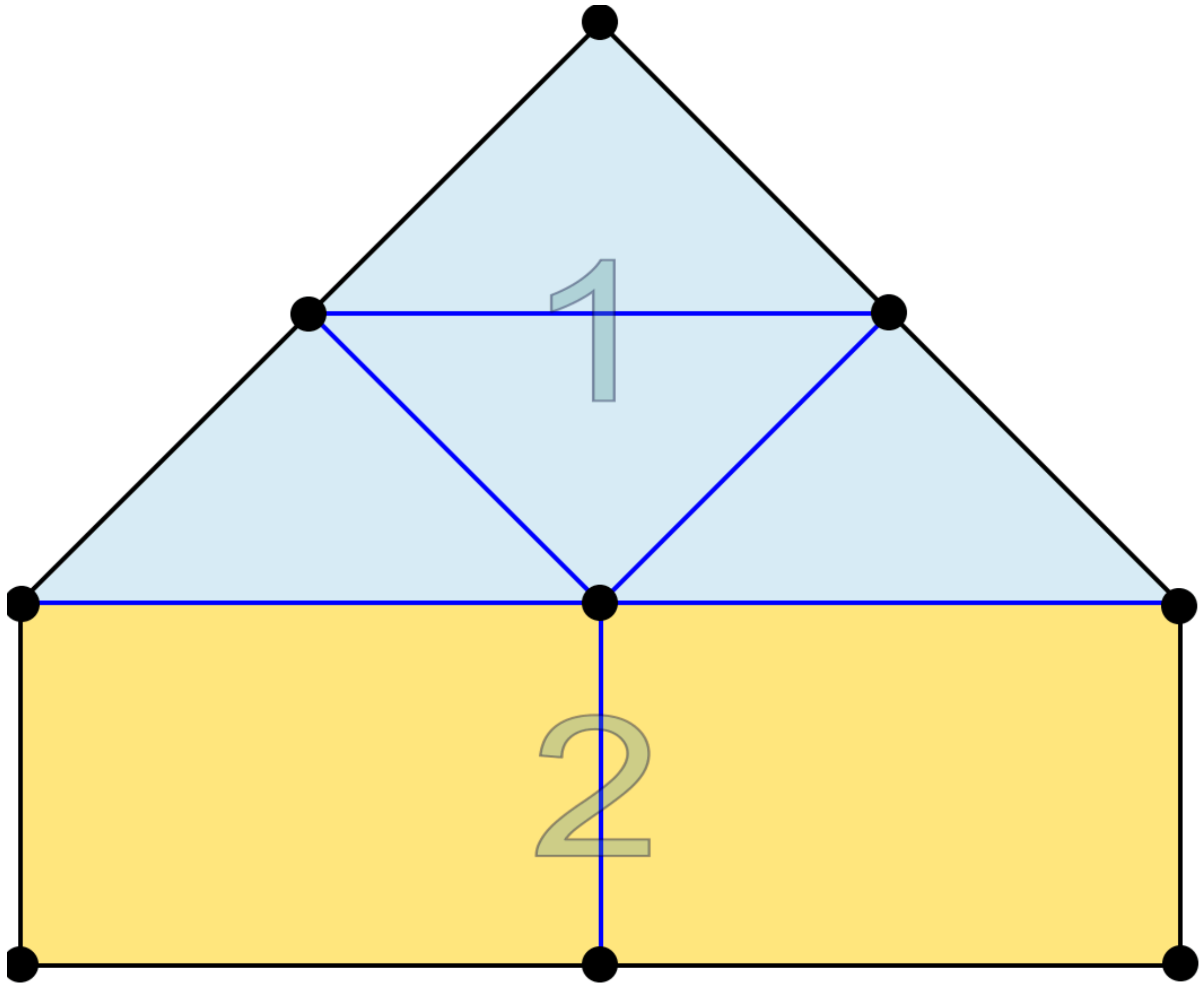


decoupling

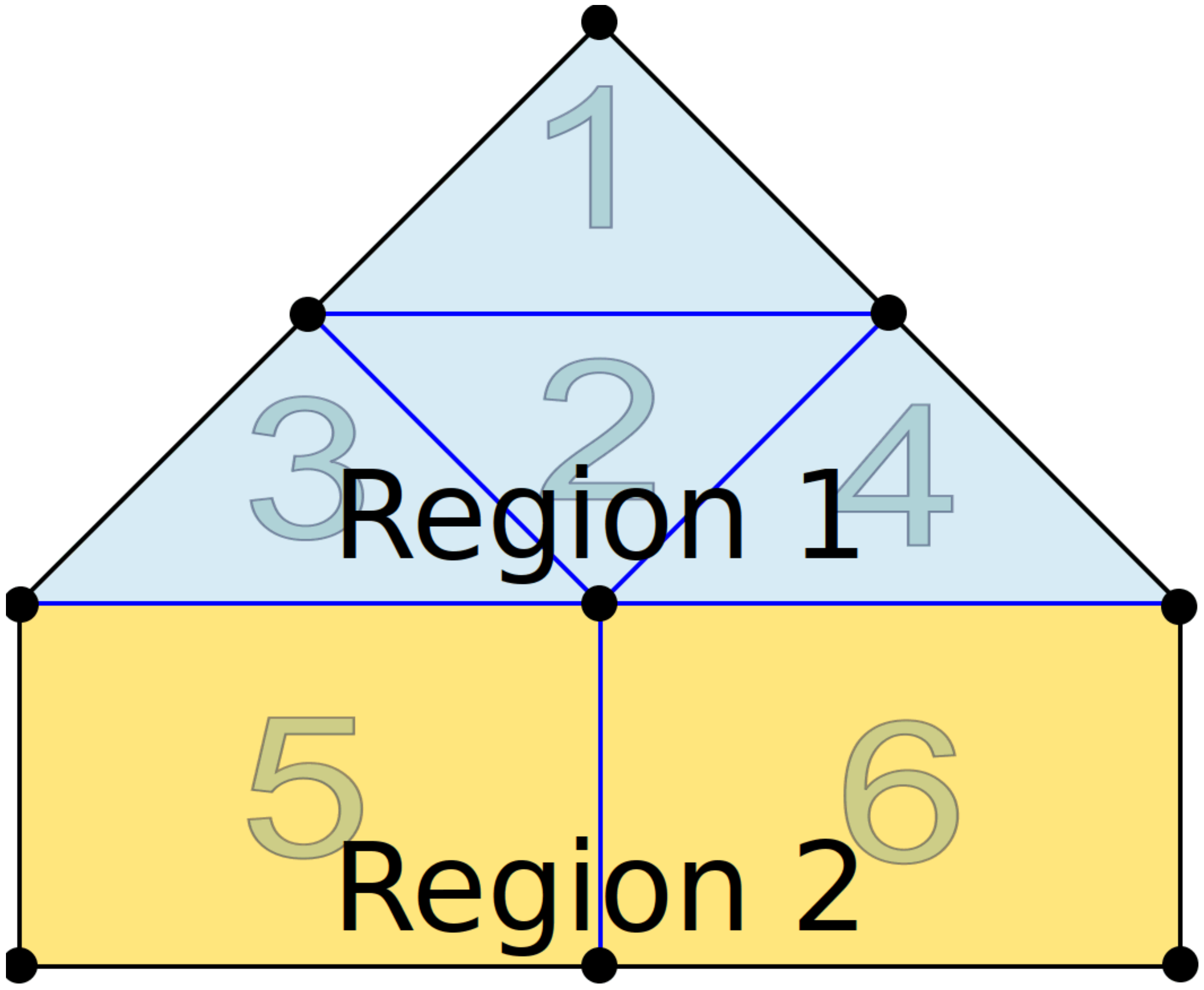


disjunct regions

Model reduction

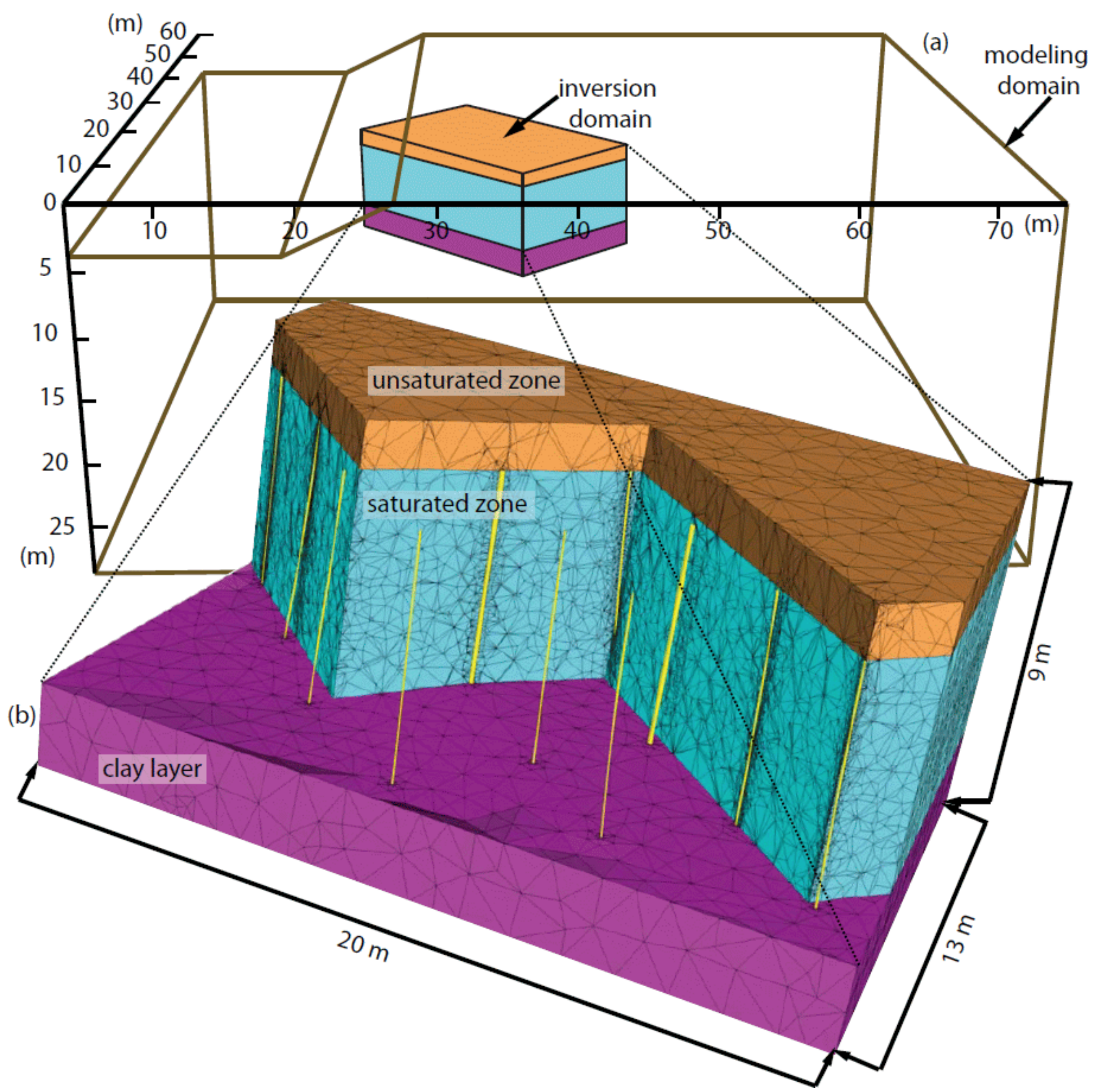


clustering models

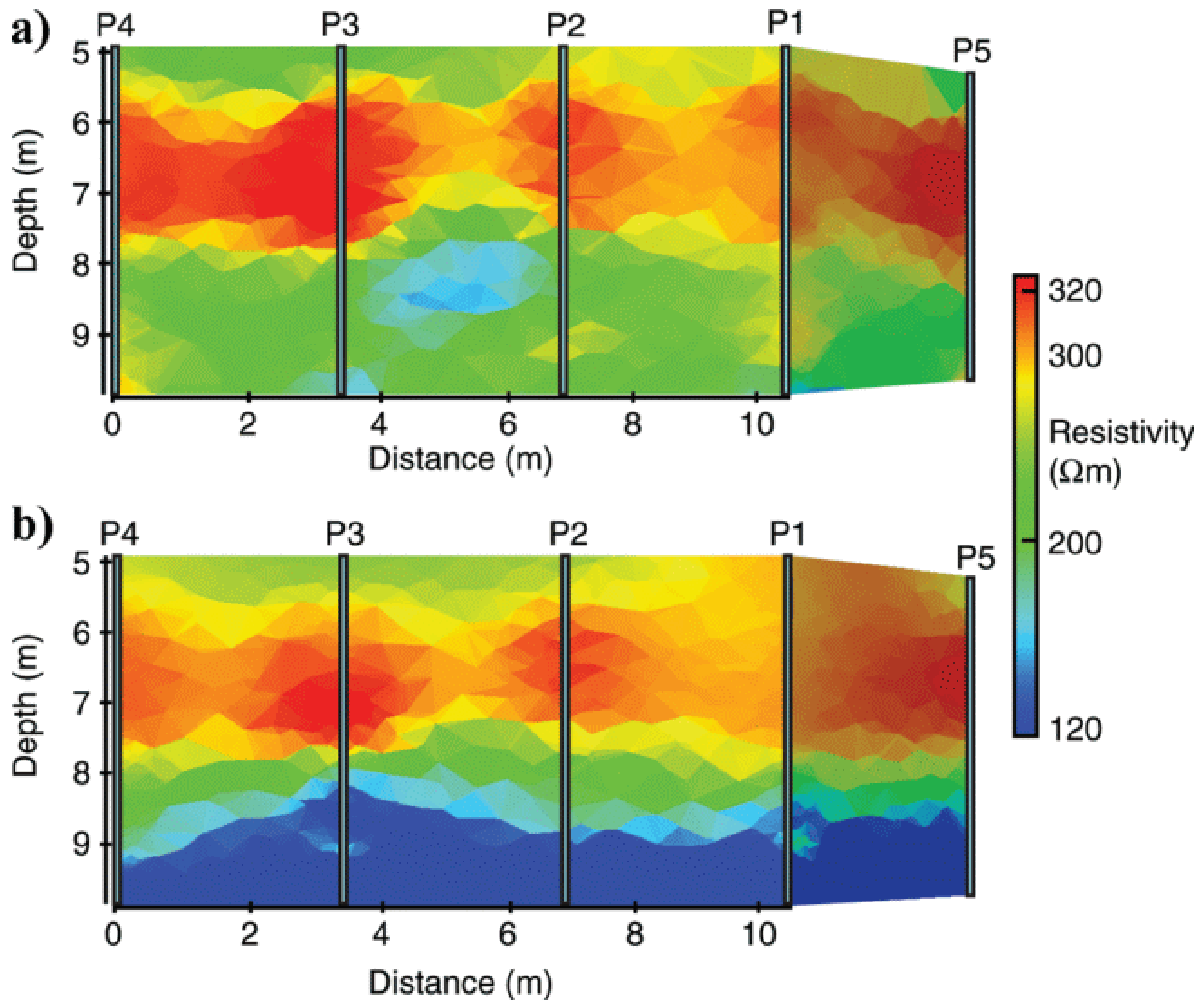


disjunct regions

Region-wise settings



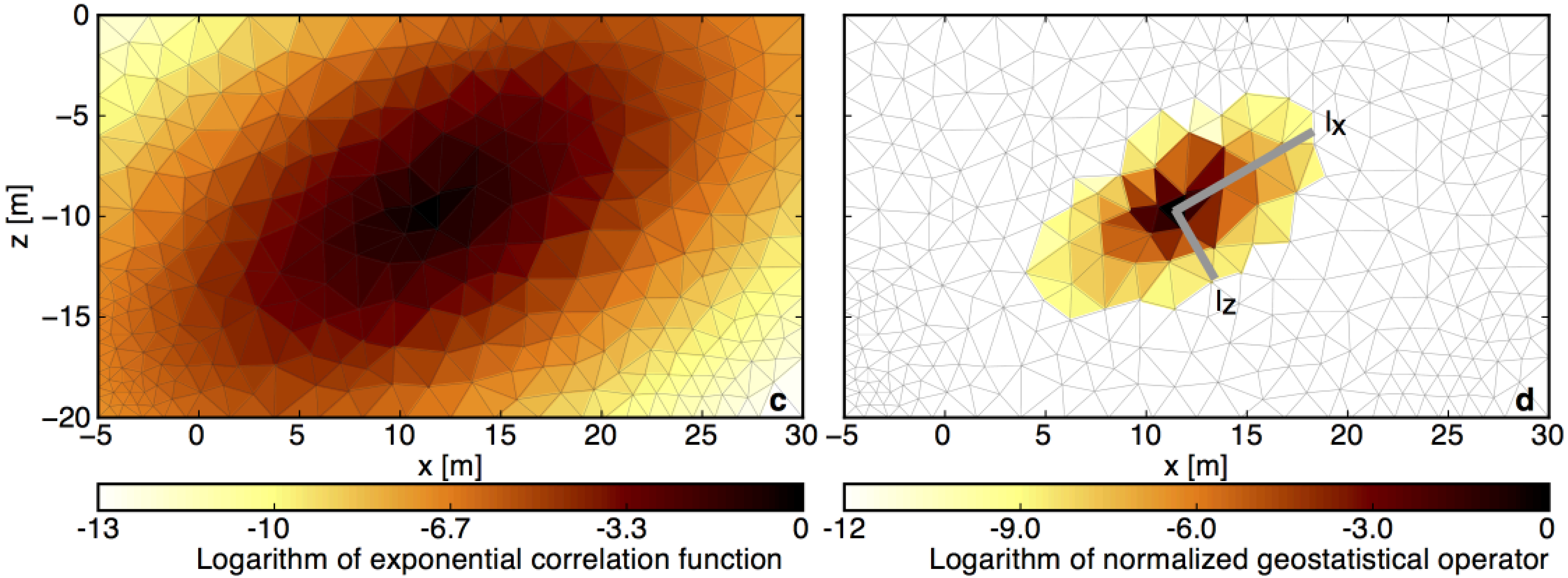
(m)
60
50
40
30
20
10
0
10
20
30
40
50
60
70 (m)
5
10
15
20
25
(m)
(a)
modeling domain
inversion domain
unsaturated zone
saturated zone
clay layer
(b)
20 m
13 m
9 m



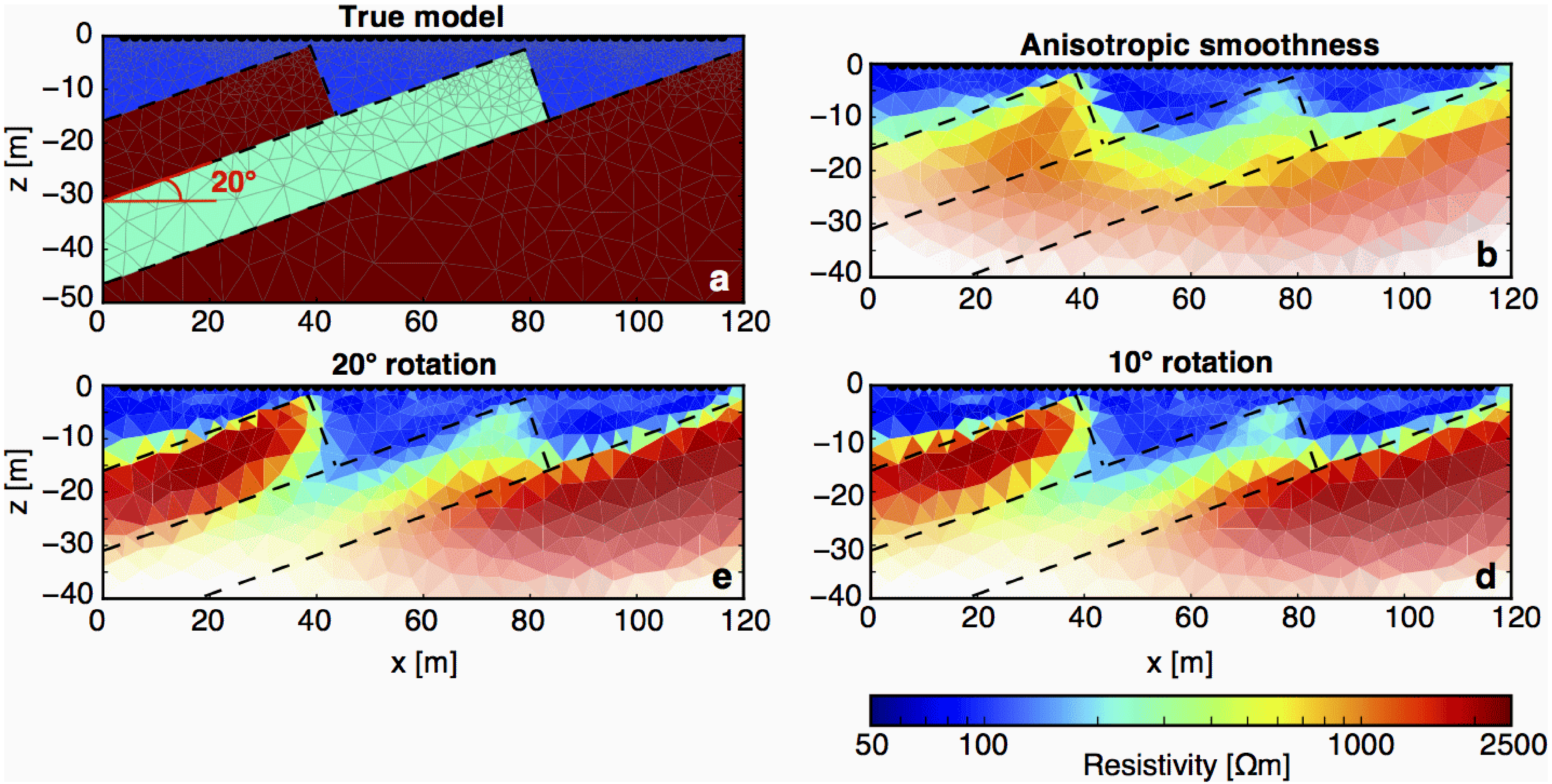
a)
b)
P4
P3
P2
P1
P5
Depth (m)
5
6
7
8
9
0
2
4
6
8
10
Distance (m)
320
300
Resistivity (Ωm)
200
120

Geostatistical regularization

$$\mathbf{M}_{ij} = \sigma^2 \exp\left(-3\sqrt{\frac{|x_i - x_j|}{I_x}^2 + \frac{|y_i - y_j|}{I_y}^2}\right) \quad \Rightarrow \quad \mathbf{C} = \mathbf{M}^{-0.5}$$



Geostatistical regularization - example



Jordi et al. (2018)

Discretization (Menke, 2012)